

VINDHYAVASINI KALWAKUNTA

(505)-328-0579 | vindhyavasini.kalwakunta@gmail.com | **GitHub:** github.com/VindhyavasiniKalwakunta

PROFESSIONAL SUMMARY

Data Engineer with 5+ years of experience in healthcare and banking specializing in Python, SQL, and AWS to build and maintain ETL pipelines. Skilled in data modeling, cleaning, and validating datasets within relational databases and data warehouses. Experienced in automating workflows using AWS Glue and Jenkins to improve operational efficiency. Collaborative in Agile environments to deliver analytics-ready data for reporting.

TECHNICAL SKILLS

- **Languages:** Python, SQL, Java
- **Data & Modeling:** Dimensional Modeling, Data Quality Rules, Data Profiling, Schema Design, Data Lineage, Logical/Physical Modeling
- **Databases:** Oracle, MySQL, PostgreSQL, Amazon Redshift, Snowflake
- **Tools:** Excel, Tableau, Git, Postman, Jira, Power BI
- **Big Data & Processing:** Apache Spark, PySpark
- **Cloud:** AWS S3, AWS Glue, AWS Lambda, AWS RDS, AWS Redshift, Snowflake
- **DevOps & Infrastructure:** Terraform, Shell Scripting, Linux Administration, Jenkins, Docker
- **Governance & Validation:** Data Reconciliation, Exception Handling, Audit & Compliance Data, Data Documentation
- **ETL / CI/CD:** AWS Glue, Jenkins, Docker, Maven
- **Operating Systems:** Linux, Windows

PROFESSIONAL EXPERIENCE

Elevance Health

Jun 2024 - Present

Data Engineer

Atlanta, GA

- Engineered end-to-end ETL pipelines using AWS Glue and Jenkins to ingest and transform structured healthcare claims data, reducing manual processing time and enabling daily analytics-ready dataset delivery.
- Developed SQL and Python transformations to integrate member and claims data across Amazon Redshift and S3, supporting HIPAA-compliant regulatory and compliance reporting workflows.
- Designed star-schema dimensional models for claims analytics, enabling downstream reporting teams to query 50M+ member records with consistent, validated data.
- Implemented data quality rule frameworks and automated validation checks to detect anomalies early in the pipeline, improving data accuracy and reducing rework for business stakeholders.
- Documented data lineage, metadata, and transformation logic across the AWS S3 data lake, strengthening governance, auditability, and cross-team data trust.
- Supported reporting and dashboard initiatives by delivering curated datasets for Tableau and Power BI reporting environments used by analytics and operations teams.

Divis Labs

Jan 2023 - Aug 2023

Data Engineer

Hyderabad, India

- Built scalable PySpark ingestion and transformation pipelines on Apache Spark for clinical and laboratory datasets, ensuring data completeness and traceability for compliance and regulatory submissions.
- Designed and optimized relational schemas in Oracle and MySQL, translating logical data models into physical structures to support efficient querying across multi-million-row clinical datasets.
- Conducted rigorous data profiling, quality checks, and reconciliation across clinical data sources, identifying and resolving discrepancies that could impact regulatory accuracy.
- Partnered with analysts, SMEs, and engineering teams to convert ambiguous business requirements into detailed technical specifications, accelerating sprint delivery in Agile workflows.
- Maintained metadata repositories, validation rule catalogs, and exception-handling documentation to reinforce data governance standards across all pipeline stages.

Cyient

Jun 2020 - Dec 2022

Data Engineer

Hyderabad, India

- Supported a banking data modernization initiative by integrating customer and transactional data from legacy systems into a centralized analytical layer, enabling unified financial reporting.
- Designed and optimized SQL pipelines and transformations for analytical workloads including balance sheet reconciliation, transaction summaries, and customer segmentation reporting.
- Executed data validation and QA checks across ingestion, transformation, and load stages, ensuring completeness, referential integrity, and lineage across financial datasets.
- Automated infrastructure deployment and operational workflows using Shell scripting, Terraform, and Jenkins to improve environment consistency and reduce manual intervention.
- Created comprehensive ETL documentation including data dictionaries, workflow diagrams, and transformation logic to support audit readiness and knowledge transfer across teams.

EDUCATION

University of Central Oklahoma

2023 - 2025

Master of Science, Computer Science

Malla Reddy Institute of Engineering and Technology

2017 - 2021

Bachelor of Technology, Computer Science Engineering

CERTIFICATIONS

- AWS Certified Developer – Associate
- Oracle Certified Java Programmer (OCPJP)
- Data Engineering with Python & SQL

PROJECTS

Rainfall Prediction Using Python

- Built a regression-based predictive model using Python (Pandas, Scikit-learn) on historical rainfall datasets, achieving strong seasonal forecasting accuracy.

Deep Learning Facial Expression Recognition Based Scoring System for Restaurants

- Developed a deep learning model to analyse customer facial expressions and automatically rate restaurant service quality, leveraging TensorFlow and OpenCV.

Smart Chatbot using OpenAI API (LLM-based System)

- Developed a client server-based AI chatbot to assist students and staff with academic and administrative queries using Large Language Models via OpenAI API.
- Implemented user authentication, chat management, FAQ storage, and chat history export features.
- Designed secure backend services to handle NLP processing, data storage, and API integration while ensuring system availability, performance, and data privacy.

Student Management Web Application (PHP & Oracle Database)

- Developed a web-based student portal using PHP and Oracle Database, implementing secure session-based authentication, role-based access control, and backend utilities for executing and handling SQL queries.
- Designed database-driven features for student profile access, login validation, and error handling using OCI functions to ensure reliable and secure data operations.